SESSION CHECK

Session 7 — The Sun, safely

Practical Astronomy Intensive

Name ______________________________ Date ________________

7 questions **20** minutes

SAFETY GATE, not a scored paper. The first three questions are pass/fail: a student who fits an eyepiece filter, trusts smoked glass, or thinks a quick glance is survivable does not touch a telescope in daylight until retaught and rechecked. Mark these before the Sun is up.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

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1 Two solar filters are on the table: one screws into the eyepiece barrel, one caps the front of the tube. Which do you fit, and why?

- a The eyepiece filter — it sits closest to the eye, which is the last line of defence
- b The eyepiece filter, with the front left open so you can still aim the scope
- c The front-aperture filter, always — it rejects the sunlight before the telescope concentrates it, so nothing downstream ever carries the heat
- d Either; both are sold as solar filters, so both are certified for the job

2 Someone offers you these for looking at the Sun: dark sunglasses, a strip of exposed photographic film, a piece of smoked glass, and a welder's glass of shade 10. The Sun looks comfortably dim through every one. Which is safe?

- a None of them — all four cut the visible glare while letting infrared through, and comfort is not the test. Only a certified solar filter (ND5 front-aperture film, or shade 14+ welding glass) is safe
- b The shade 10 welding glass — welders look at an arc through it all day
- c The exposed film and the smoked glass — these are the traditional methods and have been used for generations
- d The sunglasses, if you stack two pairs so the Sun looks properly dim

3 Set up the refractor for the Sun. Fit the front-aperture filter, deal with the finder, and hand the scope to the instructor for check — do not put your eye, or let anyone else put an eye, to the eyepiece until the instructor has personally passed it.

> *Practical — performed for your assessor.*

4 Through the filtered scope, a sunspot group shows as a dark blotch on the disc. What is a sunspot?

- a A shadow cast on the photosphere by a prominence looping above it
- b A hole in the Sun's surface, through which you are seeing down into a darker interior
- c A region where a strong magnetic field chokes convection, leaving the gas at around 4,000 K — cool only beside the 5,800 K photosphere around it; lifted out on its own it would blaze brighter than an arc lamp
- d A genuinely cold, dark patch of the Sun — cold in the ordinary sense of the word

5 We call the photosphere the Sun's 'surface' and it looks like a crisp edge through the filter. What is it actually?

- a A shell of denser gas held up by the pressure of the corona pressing down from outside
- b Just the depth at which the gas stops being opaque and light finally escapes — there is no boundary there, only a change in transparency
- c A solid or liquid surface, with the atmosphere sitting above it as on Earth
- d The top of the fusion core, where the energy is actually generated

6 With the checked filtered scope, find the sunspot groups on today's disc, sketch their positions on the blank disc template, and count the groups. Then say — out loud — whether today's count says anything about where we are in the solar cycle.

Practical — performed for your assessor.

7 The core is 15 million K. The photosphere is 5,800 K. What does the temperature do in the corona above it, and why is that interesting?

- a It holds steady at photosphere temperature, since the corona is just thinner gas at the same heat
- b It drops to near absolute zero — the corona is essentially vacuum
- c It keeps falling smoothly outwards, as you would expect moving away from the heat source
- d It jumps back up to millions of kelvin — hundreds of times hotter than the surface beneath it, which nobody has fully explained